

VPA A-Level CS — Weeks 1–9: Information Representation & Communication

AS Level Semester 1 — Weeks 1 – 9

Information Representation & Communication 信息表示与通信 (Xìnxī biǎoshì yǔ tōngxìn)

Syllabus Coverage: 1.1 Information Representation, 1.2 Communication

Paper: Paper 1 (Theory Fundamentals)

Duration: 9 weeks (Weeks 1–9)

Assessment: Diagnostic (W1), Quiz (W3), Mid-Unit Test (W6), Half-Term Exam (W9)

Week-by-Week Scheme of Work

Week 1: Number Systems — Binary, Denary, Hexadecimal 数字系统 (Shùzì xìtǒng)

Lesson 1 (40 min): Introduction to A-Level CS & Number Systems Review

Learning Objectives:

- ☐ Recall binary and denary conversion from IGCSE 复习IGCSE二进制和十进制转换
- ☐ Understand why computers use binary 理解计算机为何使用二进制
- ☐ Convert between binary, denary, and hexadecimal 二进制、十进制和十六进制之间的转换

Starter (5 min): "Do Now" — Convert 10101100 to denary. Convert 187 to binary. Self-mark using provided answers.

Main Activity (20 min):

- ☐ Teacher models: binary → denary (place value method); denary → binary (division method)
- ☐ Hexadecimal introduction: Why use hex? Compact representation of binary. Show relationship: 1 hex digit = 4 binary digits (nibble).
- ☐ Guided practice: Convert 11010111 to hex (split into nibbles: 1101 0111 = D7). Convert B4F to binary.
- ☐ Students complete conversion worksheet (Bronze/Silver/Gold)

Plenary (10 min): Mini-whiteboard challenge — teacher calls out numbers, students convert.

Homework: 10 conversion problems + vocabulary flashcards (binary, denary, hexadecimal, nibble, byte)

Differentiation:

Bronze: 8-bit conversions only; conversion tables provided

Silver: 16-bit conversions; explain WHY hex is used

Gold: 32-bit conversions; research octal system and compare with hex

EAL Support:

- ☐ Visual place-value grid (128, 64, 32, 16, 8, 4, 2, 1) displayed permanently
- ☐ Colour-code: binary (blue), denary (green), hex (red)

☐ Cognates: "binary" → 二进制 (èr jìnzhì), "decimal" → 十进制 (shí jìnzhì)

☐ Sentence starter: "To convert binary to denary, I multiply each digit by its place value and add the results."
将二进制转换为十进制时, 我将每个数字乘以其位权然后相加。

Lesson 2 (40 min): Two's Complement & Negative Numbers 补码与负数 (Bǔmǎ yǔ fùshù)

Learning Objectives:

- ☐ Represent negative integers using two's complement 使用补码表示负整数
- ☐ Perform binary addition and subtraction 进行二进制加减法
- ☐ Understand range limitations of fixed-bit representations 理解固定位数表示的范围限制

Starter (5 min): How would you represent -5 in binary? Discuss in pairs.

Main Activity (20 min):

☐ Sign-magnitude: leftmost bit = sign (0 positive, 1 negative). Problem: two representations of zero.
原码表示法: 最高位为符号位, 但存在正零和负零的问题。

☐ Two's complement method: Invert all bits, add 1. Demonstrate with 8-bit: +5 = 00000101, -5 = 11111011.

☐ Range of 8-bit two's complement: -128 to +127. 8位补码范围: -128到+127

☐ Binary addition demo: $7 + (-3) = 4$. Show that subtraction = addition of two's complement.

☐ Overflow: What happens when $127 + 1$ in 8-bit? Wraps to -128!

Plenary (10 min): Complete truth table: show +5, -5, +127, -128 in sign-magnitude and two's complement.

Homework: Two's complement worksheet (15 problems) + write explanation of overflow in own words

Common Misconceptions:

- ☐ ☐ "The leftmost bit is just a sign bit" — In two's complement, it has negative place value (-128 for 8-bit)
- ☐ ☐ "Two's complement and sign-magnitude are the same" — Clarify both methods and why two's complement is preferred
- ☐ ☐ "You can represent any number in 8 bits" — Emphasise range limits; introduce concept of overflow

Differentiation:

Bronze: 4-bit two's complement only; step-by-step guide provided

Silver: 8-bit; identify overflow conditions

Gold: 16-bit; prove mathematically why two's complement works (modular arithmetic)

Lesson 3 (40 min): Binary Coded Decimal (BCD) & Real-World Applications BCD编码与实际应用

Learning Objectives:

- ☐ Convert denary to BCD and vice versa 十进制与BCD码互相转换
- ☐ Compare BCD with pure binary for specific applications 比较BCD与纯二进制在特定应用中的优劣
- ☐ Understand when each representation is appropriate 理解每种表示法的适用场景

Starter (5 min): Why do digital clocks display digits directly rather than calculating from binary?

Main Activity (20 min):

☐ BCD: Each decimal digit = 4 bits. 59 = 0101 1001. BCD: 每个十进制数字用4位二进制表示。

☐ Applications: Digital clocks, calculators, financial systems (avoids rounding errors), embedded displays.

☐ Comparison: BCD is human-readable but inefficient (wastes bit patterns 1010-1111). Pure binary is compact but requires conversion for display.

☐ Group task: Design a digital clock display circuit using BCD. Sketch 7-segment display connection.

Plenary (10 min): Exam-style question: "A cash register stores prices. Explain why BCD might be preferred over pure binary." (2 marks)

Homework: Research: Find 3 real devices that use BCD. Explain why.

Assessment: Exit ticket — Convert 947 to BCD. Compare storage needed for 947 in BCD vs 16-bit binary.

Week 1 Assessment: Baseline Diagnostic (30 minutes)

- ☐ 10 MCQs: Binary/hex/denary conversions (IGCSE level)
- ☐ 3 short-answer: Explain why computers use binary; Convert with working; BCD application
- ☐ Total: 30 marks
- ☐ Purpose: Identify students needing intervention; group for differentiation
- ☐ 诊断测试目的: 识别需要干预的学生, 进行分层分组

Week 2: Text, Images, and Sound Representation 文本、图像和声音表示 (Wénběn, túxiàng hé shēngyīn biǎoshì)

Lesson 1 (40 min): Text Encoding — ASCII & Unicode 文本编码: ASCII与Unicode

Learning Objectives:

- ☐ Explain how characters are encoded as binary 解释字符如何编码为二进制
- ☐ Compare ASCII and Unicode 比较ASCII和Unicode
- ☐ Calculate storage requirements for text files 计算文本文件的存储需求

Starter (5 min): Type your name in Notepad. Save as different encodings. Compare file sizes.

Main Activity (20 min):

- ☐ ASCII: 7 bits, 128 characters. Extended ASCII: 8 bits, 256 characters. Show ASCII table.
- ☐ Problem: ASCII cannot represent Chinese characters. ASCII无法表示中文字符。Enter Unicode.
- ☐ Unicode: UTF-8 (variable width, 1–4 bytes), UTF-16 (2 or 4 bytes), UTF-32 (fixed 4 bytes).
- ☐ Storage calculation: "HELLO" in ASCII = 5 bytes. "你好" in UTF-8 = 6 bytes (3 bytes per CJK character).
- ☐ Exam tip: CIE expects knowledge of Unicode but not detailed UTF encoding rules.

Activity: Students encode their full name in ASCII (English letters) and discuss how Unicode enables their Chinese name.

Plenary (10 min): "Why is UTF-8 dominant on the web?" Discuss in pairs. Answer: Backward compatible with ASCII, efficient for English, supports all languages.

Homework: Calculate storage for: (a) 500-character English essay in ASCII, (b) Same essay with 50 Chinese characters in UTF-8.

Common Misconceptions:

- ☐ ☐ "ASCII and Unicode are the same thing" — Clarify ASCII is subset of Unicode
- ☐ ☐ "One character always = one byte" — Emphasise variable-width encoding

Bilingual Keywords:

- ☐ Character encoding 字符编码 (zìfú biānmǎ)
- ☐ Code point 码点 (mǎdiǎn)
- ☐ Character set 字符集 (zìfú jí)

Lesson 2 (40 min): Image Representation — Bitmap vs Vector 图像表示: 位图与矢量图

Learning Objectives:

- ☐ Explain how bitmap images are stored 解释位图图像的存储方式
- ☐ Calculate bitmap file size from dimensions and colour depth 根据尺寸和色深计算位图文件大小
- ☐ Compare bitmap and vector graphics 比较位图和矢量图

Starter (5 min): Display two identical-looking images: one 5MB bitmap, one 50KB vector. Ask: Why the difference?

Main Activity (20 min):

- ☐ Bitmap = grid of pixels. Each pixel = colour value.
- ☐ Colour depth: 1-bit (black/white), 8-bit (256 colours), 24-bit (16.7 million colours = true colour)
- ☐ File size formula: Width $\times$ Height $\times$ Colour depth (in bits), then convert to bytes/KB/MB
- ☐ Worked example: 1024×768 image at 24-bit colour = $1024 \times 768 \times 24 = 18,874,368$ bits = 2.25 MB
- ☐ Metadata: Resolution, colour depth, dimensions all affect file size.
- ☐ Vector: Mathematical descriptions of shapes (lines, curves, polygons). Scalable without quality loss.
- ☐ Comparison table: File size, scalability, editing, best use cases.

Practical Activity (10 min): Students zoom into bitmap and vector versions of same logo. Observe pixelation vs smooth scaling.

Plenary (5 min): Exam-style: "A photographer stores images. Justify bitmap over vector." (2 marks)

Homework: Calculate file sizes for: (a) 1920×1080 at 32-bit, (b) 4000×3000 at 24-bit. Research: When would a graphic designer use SVG?

Differentiation:

Bronze: Fill-in file size formula; calculate with given dimensions

Silver: Calculate from scratch; explain resolution vs file size relationship

Gold: Calculate for 4K video frame (3840×2160 at 24-bit, 60fps); discuss bandwidth requirements

Lesson 3 (40 min): Sound Representation — Sampling & Bit Depth 声音表示：采样与位深度

Learning Objectives:

- ☐ Explain how analogue sound is digitised 解释模拟声音如何数字化
- ☐ Define sample rate and bit depth 定义采样率和位深度
- ☐ Calculate audio file sizes 计算音频文件大小
- ☐ Discuss trade-offs in audio quality vs file size 讨论音质与文件大小的权衡

Starter (5 min): Listen to same song: 64 kbps MP3 vs 320 kbps MP3 vs WAV. Can you tell the difference?

Main Activity (20 min):

- ☐ ADC process: Sample $\rightarrow$ Quantise $\rightarrow$ Encode. Show waveform diagram.
- ☐ Sample rate: Samples per second (Hz). CD quality = 44.1 kHz. Nyquist theorem: Sample at $2 \times$ maximum frequency.
- ☐ Bit depth: Bits per sample. CD = 16-bit. Studio = 24-bit. Higher = more dynamic range.
- ☐ File size formula: Sample rate $\times$ Bit depth $\times$ Channels $\times$ Duration (seconds)
- ☐ Worked example: $44,100 \text{ Hz} \times 16\text{-bit} \times 2 \text{ channels} \times 180 \text{ seconds} = 31,752,000 \text{ bytes} \approx 30.3 \text{ MB}$
- ☐ Compression: Lossy (MP3, AAC — psychoacoustic models) vs Lossless (FLAC, ALAC)

Activity (10 min): Calculate file sizes for: (a) 3-minute mono speech at 22 kHz, 8-bit, (b) 5-minute stereo music at 96 kHz, 24-bit.

Plenary (5 min): "Why do streaming services use lossy compression?" Discuss quality vs bandwidth trade-off.

Homework: Research: What sample rate and bit depth does your phone use for voice calls? Why?

Bilingual Keywords:

- ☐ Sample rate 采样率 (cǎiyàng lǜ)
- ☐ Bit depth 位深度 (wèi shēndù)
- ☐ Analogue-to-digital converter 模数转换器 (mó-shù zhuǎnhuàn qì)
- ☐ Quantisation 量化 (liàng huà)
- ☐ Nyquist theorem 奈奎斯特定理 (Nài kuí sī tè dìng lǐ)

Week 3: Compression 数据压缩 (Shùjù yāsuō)

Lesson 1 (40 min): Lossy vs Lossless Compression 有损与无损压缩

Learning Objectives:

- ☐ Distinguish between lossy and lossless compression 区分有损和无损压缩
- ☐ Give examples of each type 举例说明每种类型
- ☐ Explain when each type is appropriate 解释何时使用每种类型

Starter (5 min): Show identical image saved as BMP, PNG, JPEG. File sizes: 2.25 MB, 850 KB, 120 KB. Why different?

Main Activity (20 min):

- ☐ Lossless: No data lost. Original can be perfectly reconstructed. Examples: PNG, GIF, ZIP, FLAC, RLE.
- ☐ Lossy: Data permanently removed. Smaller files but quality loss. Examples: JPEG, MP3, MPEG.
- ☐ Key distinction: Lossy uses perceptual models (human eye less sensitive to colour detail than brightness; ear less sensitive to quiet sounds masked by loud ones).
- ☐ Use cases: Text/docs = always lossless. Photos = lossy acceptable. Medical images = lossless. Music archiving = lossless; streaming = lossy.
- ☐ Group discussion: "Should NASA use lossy compression for Mars rover images?" NASA应该对火星车图像使用有损压缩吗?

Plenary (10 min): Categorise: BMP, MP3, FLAC, PNG, JPEG, ZIP, GIF, MPEG. Lossy or lossless?

Homework: Create decision tree: "Which compression should I use?" with 5+ scenarios.

Common Misconceptions:

- ☐ ☐ "Lossy compression damages the original file" — Clarify: compressed copy is separate; original remains
- ☐ ☐ "ZIP is lossy" — Emphasise ZIP is always lossless

Lesson 2 (40 min): Run-Length Encoding (RLE) 行程长度编码

Learning Objectives:

- ☐ Encode data using RLE 使用RLE编码数据
- ☐ Decode RLE data 解码RLE数据
- ☐ Evaluate effectiveness of RLE for different data types 评估RLE对不同数据类型的有效性

Starter (5 min): String: AAAAABBBCCDAAAA. How would you compress this?

Main Activity (20 min):

- ☐ RLE principle: Replace consecutive identical symbols with count + symbol.
- ☐ Example: AAAAABBBCCDAAAA → 5A3B2C1D4A
- ☐ Worked examples with binary: 0000111111000001111 → 0405105040 (using delimiter) or 4 0 5 1 5 0 4 1
- ☐ Limitations: What if data has no runs? ABCABCABC → 1A1B1C1A1B1C1A1B1C (expands!) 如果数据没有连续重复, RLE会膨胀!
- ☐ Best for: Simple graphics, black-and-white images, fax transmissions.
- ☐ Students practise encoding/decoding with worksheet.

Plenary (10 min): "RLE reduced a file from 1000 bytes to 400 bytes. The original was mostly black space. Explain why RLE worked well." (3 marks)

Homework: Create your own RLE encoding for a simple 10×10 pixel black-and-white image.

Differentiation:

Bronze: RLE on text strings with clear runs

Silver: RLE on binary image data; identify when RLE fails

Gold: Design improved RLE variant with escape codes for raw data

Lesson 3 (40 min): Dictionary-Based Compression (Huffman Coding Introduction) 字典压缩与霍夫曼编码

Learning Objectives:

- ☐ Explain dictionary-based compression principles 解释字典压缩原理
- ☐ Understand basic Huffman coding concept 理解霍夫曼编码基本概念
- ☐ Compare RLE and dictionary methods 比较RLE和字典方法

Starter (5 min): Text: "the quick brown fox jumps over the lazy dog the quick brown fox..." Notice repetition?

Main Activity (20 min):

- ☐ Dictionary compression: Build table of recurring patterns; replace with shorter codes. LZ77/LZ78 family (ZIP uses this).
- ☐ Huffman coding: Variable-length codes. Frequent symbols = shorter codes. 频率高的符号用更短的编码。
- ☐ Example: Text with many 'e's and few 'z's. 'e' = 01, 'z' = 11110.
- ☐ Prefix-free property: No code is prefix of another. Ensures unambiguous decoding.
- ☐ Show Huffman tree construction (simplified — CIE requires understanding, not full construction).
- ☐ Comparison: RLE (simple, fast, limited use) vs Huffman (complex, general-purpose, better ratios).

Activity (10 min): Given frequency table, assign Huffman codes. Calculate compression ratio.

Plenary (5 min): "Why can't Huffman coding use fixed-length codes for all symbols?"

Homework: Research: How does JPEG combine lossy and lossless compression? (DCT + Huffman)

Week 3 Assessment: Information Representation Quiz (20 minutes)

- ☐ Q1-3: Number conversions (9 marks)
- ☐ Q4: Two's complement (4 marks)
- ☐ Q5: Text encoding comparison (4 marks)
- ☐ Q6: Image file size calculation (6 marks)
- ☐ Q7: Audio file size calculation (6 marks)
- ☐ Q8: Compression method selection (6 marks)
- ☐ Total: 35 marks — Formative, self-marked with teacher review

Week 4: Communication — Networks & Client-Server 通信：网络与客户端-服务器 (Tōngxìn: Wǎngluò yǔ kèhùduān-fúwùqì)

Lesson 1 (40 min): Network Types & Topologies 网络类型与拓扑结构

Learning Objectives:

- ☐ Define LAN, WAN, PAN, MAN 定义局域网、广域网、个人网、城域网
- ☐ Describe client-server and peer-to-peer models 描述客户端-服务器和对等网络模型
- ☐ Explain advantages and disadvantages of each model 解释每种模型的优缺点

Starter (5 min): How do you share files with friends? WeChat? Baidu Netdisk? School server? What's the difference?

Main Activity (20 min):

- ☐ LAN: Local Area Network — school, home, office. High speed, private, limited range. 局域网：学校、家庭、办公室
- ☐ WAN: Wide Area Network — spans cities/countries. Internet is largest WAN. Uses leased lines, satellites, fibre.
- ☐ PAN: Personal Area Network — Bluetooth devices around one person.
- ☐ MAN: Metropolitan Area Network — city-wide (e.g., municipal WiFi).

☐ Client-server: Central server provides services; clients request. Examples: school file server, web server, email server.

— Advantages: Centralised security, backups, administration

— Disadvantages: Single point of failure, expensive server hardware

☐ Peer-to-peer (P2P): No central server; all nodes equal. Examples: BitTorrent, blockchain, early Skype.

— Advantages: No single point of failure, scalable, low cost

— Disadvantages: Security harder, no central backup, inconsistent performance

Activity (10 min): Scenario cards: "Design network for..." (a) small office, (b) multinational corporation, (c) gaming tournament. Students justify choices.

Plenary (5 min): "Why does your school use client-server for files but P2P might be better for video calls?"

Homework: Draw network diagram for your home. Label LAN/WAN boundaries, identify client-server vs P2P examples.

Bilingual Keywords:

☐ Network topology 网络拓扑 (wǎngluò tuòpū)

☐ Node 节点 (jiédiǎn)

☐ Bandwidth 带宽 (dàikuān)

☐ Latency 延迟 (yánchí)

Lesson 2 (40 min): Network Hardware 网络硬件设备

Learning Objectives:

☐ Identify and explain functions of network hardware 识别并解释网络硬件的功能

☐ Describe how data flows through network devices 描述数据如何流经网络设备

☐ Compare switches, routers, and hubs 比较交换机、路由器和集线器

Starter (5 min): Show images of hub, switch, router. Can students identify? What's the difference?

Main Activity (20 min):

☐ Hub (集线器 jíxiànrì): Layer 1 device. Broadcasts all data to all ports. Cheap but inefficient. Obsolete.

☐ Switch (交换机 jiāohuànqī): Layer 2 device. Learns MAC addresses, sends data only to correct port. Efficient, used in LANs.

☐ Router (路由器 lùróuqī): Layer 3 device. Connects networks, uses IP addresses, makes forwarding decisions. Gateway to Internet.

☐ NIC (网卡 wǎngkǎ): Network Interface Card — connects device to network. Has unique MAC address.

☐ Access Point (接入点 jiērù diǎn): Wireless connection point for WiFi devices.

☐ Modem (调制解调器 tiáozhì jiětiáo qì): Modulates/demodulates signals for transmission over telephone lines/cable.

☐ Data flow exercise: Trace path from your phone to www.google.com. Phone → WiFi AP → Switch → Router → ISP → Google server.

Plenary (10 min): Exam question: "Explain why a switch is more efficient than a hub in a school LAN." (3 marks)

Homework: Label blank network diagram with correct devices. Write function of each in English + Chinese.

Common Misconceptions:

☐ "Router and switch are the same" — Clarify layer difference (2 vs 3) and addressing (MAC vs IP)

☐ "WiFi and Internet are the same" — WiFi = wireless LAN access; Internet = global WAN

Lesson 3 (40 min): Protocols & Packet Switching 协议与分组交换

Learning Objectives:

☐ Define protocol and explain why protocols are needed 定义协议并解释为何需要协议

☐ Describe the packet switching process 描述分组交换过程

☐ Explain advantages of packet switching over circuit switching 解释分组交换相对电路交换的优势

Starter (5 min): "If I shout 'HELLO' in this room, everyone hears. If I send a letter, only recipient gets it. How does the Internet decide?"

Main Activity (20 min):

☐ Protocol: Set of rules governing data communication. Without protocols, devices couldn't understand each other. 协议是数据通信的规则集。

☐ Packet switching: Data split into packets; each packet travels independently; reassembled at destination.

☐ Packet structure: Header (source IP, destination IP, sequence number, error checking) + Payload (data) + Trailer.

☐ Analogy: Sending a book by tearing pages into envelopes, each addressed and numbered. Recipient reassembles by page numbers.

☐ Advantages: Efficient bandwidth use, resilient (packets can reroute), multiple users share lines.

☐ Circuit switching: Dedicated path (old telephone system). Reliable but wastes bandwidth when silent.

☐ Exam tip: Always mention "packets can take different routes" and "resilience/reliability" for packet switching questions.

Activity (10 min): Simulation — Students = routers. Teacher sends "packets" (index cards with numbers) through classroom network. Some cards take different paths. Reassemble message at destination.

Plenary (5 min): "Why is packet switching essential for video streaming?" Discuss: shared bandwidth, error recovery, scalability.

Homework: Explain packet switching to a family member. Write their questions and your answers.

Bilingual Keywords:

☐ Packet 分组/数据包 (fēnzǔ/shùjùbāo)

☐ Header 首部/头部 (shǒubù/tóubù)

☐ Payload 有效载荷 (yǒuxiào zàihè)

☐ Circuit switching 电路交换 (diànlù jiāohuàn)

☐ Packet switching 分组交换 (fēnzǔ jiāohuàn)

Week 5: Internet Technologies — WWW, Email, FTP

互联网技术：万维网、电子邮件、文件传输 (Hùliánwǎng jìshù)

Lesson 1 (40 min): World Wide Web & URLs 万维网与URL

Learning Objectives:

☐ Distinguish between Internet and WWW 区分互联网和万维网

☐ Explain how URLs work 解释URL的工作原理

☐ Describe roles of web browser and web server 描述浏览器和服务器的作用

Starter (5 min): "The Internet and the World Wide Web are the same thing." True or false? Vote with hands.

Main Activity (20 min):

☐ Internet = global network of interconnected networks (infrastructure). 互联网是全球互联网络的基础设施。

☐ WWW = service running ON the Internet (information system using HTTP). 万维网是运行在互联网上的信息服务。

☐ Other Internet services: Email, FTP, VoIP, streaming — none of these are "the WWW."

☐ URL structure: protocol://domain:port/path?query#fragment

— Example: <https://www.vpa.edu.cn:443/admissions?year=2024#fees>

— Protocol (https), domain (www.vpa.edu.cn), port (443 default for HTTPS), path (/admissions), query (?year=2024), fragment (#fees)

- ☐ Web browser: Requests resources using HTTP, renders HTML/CSS/JavaScript.
- ☐ Web server: Listens for requests, serves files, processes dynamic content (PHP, Python, etc.).
- ☐ HTTP request/response cycle: Browser sends GET request → Server responds with status code (200 OK, 404 Not Found, 500 Server Error) + HTML.

Activity (10 min): Students dissect 5 real URLs. Identify protocol, domain, path, query parameters.

Plenary (5 min): "Why might a URL contain a query string? Give two examples." (2 marks)

Homework: Research status codes. List 5 common codes and what they mean. 研究HTTP状态码

Lesson 2 (40 min): Email & FTP 电子邮件与文件传输

Learning Objectives:

- ☐ Explain how email is sent and received 解释电子邮件的发送和接收
- ☐ Describe protocols: SMTP, POP3, IMAP 描述SMTP、POP3、IMAP协议
- ☐ Explain FTP and its purpose 解释FTP及其用途

Starter (5 min): When you send an email from VPA to a friend in UK, what happens? Brainstorm in pairs.

Main Activity (20 min):

- ☐ Email architecture: MUA (Mail User Agent) → MSA → MTA → MDA → MUA. Show diagram.
- ☐ SMTP (Simple Mail Transfer Protocol): Sends email FROM client TO server, server TO server. Port 25/587. SMTP用于发送邮件。
- ☐ POP3 (Post Office Protocol v3): Downloads emails to single device; deletes from server. Port 110. POP3下载邮件到单一设备。
- ☐ IMAP (Internet Message Access Protocol): Synchronises across devices; leaves copy on server. Port 143. IMAP跨设备同步。
- ☐ Comparison: POP3 good for single device, limited storage. IMAP good for multiple devices, cloud access.
- ☐ FTP (File Transfer Protocol): Transfers files between client and server. Uses separate control (port 21) and data (port 20) connections. FTP用于客户端和服务端之间传输文件。
- ☐ FTP vs HTTP: FTP designed for file transfer, supports upload/download/resume. HTTP designed for web pages.
- ☐ Security: FTP sends passwords in plaintext. SFTP (SSH File Transfer Protocol) encrypts. FTP明文传输密码, SFTP加密传输。

Activity (10 min): Scenario: Design email setup for VPA staff. Should they use POP3 or IMAP? Justify.

Plenary (5 min): Exam question: "Explain why IMAP is preferred over POP3 for users with multiple devices." (3 marks)

Homework: Create comparison table: SMTP, POP3, IMAP, FTP — protocol, port, purpose, advantages, disadvantages.

Bilingual Keywords:

- ☐ SMTP 简单邮件传输协议 (jiǎndān yóujiàn chuánshū xiéyì)
- ☐ POP3 邮局协议第三版 (yóujú xiéyì dì sān bǎn)
- ☐ IMAP 互联网消息访问协议 (hùliánwǎng xiāoxī fǎngwèn xiéyì)
- ☐ FTP 文件传输协议 (wénjiàn chuánshū xiéyì)

Lesson 3 (40 min): HTML & CSS Basics (CIE Context) HTML与CSS基础

Learning Objectives:

- ☐ Understand purpose of HTML and CSS 理解HTML和CSS的用途
- ☐ Identify basic HTML tags 识别基本HTML标签
- ☐ Explain relationship between HTML, CSS, and browser rendering 解释HTML、CSS与浏览器渲染的关系

Starter (5 min): Right-click any webpage → "View page source." What do you see?

Main Activity (20 min):

- ☐ HTML (HyperText Markup Language): Structures content. Tags define elements: `<html>`, `<head>`, `<body>`, `<h1>`, `<h2>`, `<p>`, `<a>`, `<img>`, `<table>`, `<div>`, `<span>`.
- ☐ CSS (Cascading Style Sheets): Controls presentation. Colours, fonts, layout, responsiveness.
- ☐ Separation of concerns: HTML = structure, CSS = style, JavaScript = behaviour.
- ☐ Browser rendering: Parse HTML → Build DOM → Parse CSS → Build CSSOM → Combine → Render tree → Layout → Paint.
- ☐ CIE requirement: Students need to recognise HTML/CSS, not write complex pages. Focus on understanding role, not syntax mastery.
- ☐ Practical: Students modify a simple HTML file (change title, add paragraph, change colour via CSS).

Plenary (10 min): "Why is separating HTML and CSS considered good practice?" Discuss maintainability, accessibility, responsive design.

Homework: Create a simple HTML page about yourself with at least 5 different tags. Add CSS to change background colour.

Differentiation:

Bronze: Modify existing HTML file; identify 5 tags

Silver: Create HTML page from scratch; add inline CSS

Gold: Create page with external CSS file; add class selectors

Week 6: TCP/IP Stack, IP Addressing & DNS TCP/IP协议栈、IP地址与DNS

Lesson 1 (40 min): The TCP/IP Protocol Stack TCP/IP协议栈

Learning Objectives:

- ☐ Describe the four layers of TCP/IP 描述TCP/IP的四层结构
- ☐ Explain function of each layer 解释每层的功能
- ☐ Compare TCP/IP with OSI model 比较TCP/IP与OSI模型

Starter (5 min): Sending a letter: write letter → put in envelope → add address → post office sorts → delivery. How is this like Internet communication?

Main Activity (20 min):

- ☐ TCP/IP = suite of protocols organised in layers. Each layer has specific function.
- ☐ Application Layer: User-facing protocols (HTTP, FTP, SMTP, POP3, DNS). 应用层: 用户直接使用的协议。
- ☐ Transport Layer: TCP (reliable, connection-oriented, error checking) vs UDP (fast, connectionless, no error checking). 传输层: TCP可靠传输, UDP快速传输。
- ☐ Internet/Network Layer: IP addressing, routing, packet forwarding. 网络层: IP地址和路由。
- ☐ Link/Network Access Layer: Physical transmission (Ethernet, WiFi, fibre). MAC addressing. 链路层: 物理传输。
- ☐ Encapsulation: Data passes down stack; each layer adds header. At destination, headers stripped (decapsulation).
- ☐ OSI model: 7 layers (Physical, Data Link, Network, Transport, Session, Presentation, Application). TCP/IP merged some OSI layers.
- ☐ Exam mnemonic: "All People Seem To Need Data Processing" (OSI, bottom-up)

Activity (10 min): Students label blank diagram of TCP/IP stack. Add example protocols to each layer.

Plenary (5 min): "Why does the Internet use layered protocols?" Discuss: modularity, interoperability, troubleshooting, standardisation.

Homework: Create TCP/IP stack poster with protocols, functions, and analogies.

Common Misconceptions:

- ☐ ☐ "TCP/IP is one protocol" — It's a suite of protocols
- ☐ ☐ "TCP and IP are the same thing" — TCP = transport layer (reliability); IP = network layer (addressing/routing)

Lesson 2 (40 min): IP Addressing — IPv4 & IPv6 IP地址: IPv4与IPv6

Learning Objectives:

- ☐ Explain structure of IPv4 address 解释IPv4地址结构
- ☐ Understand subnet mask and private IP ranges 理解子网掩码和私有IP范围
- ☐ Explain need for IPv6 解释IPv6的必要性

Starter (5 min): How many devices in your home connect to Internet? Each needs unique address. Is that a problem?

Main Activity (20 min):

- ☐ IPv4: 32-bit address, dotted decimal notation (e.g., 192.168.1.1). IPv4: 32位地址, 点分十进制表示。
- ☐ Each octet = 8 bits, range 0–255. Total addresses: $2^{32} \approx 4.3$ billion.
- ☐ Problem: 4.3 billion IPv4地址耗尽!
- ☐ Private IP ranges (RFC 1918): 10.0.0.0–10.255.255.255, 172.16.0.0–172.31.255.255, 192.168.0.0–192.168.255.255
- ☐ NAT (Network Address Translation): Multiple devices share one public IP. Router translates.
- ☐ IPv6: 128-bit address, hexadecimal notation (e.g., 2001:0db8:85a3::8a2e:0370:7334). IPv6: 128位地址, 十六进制表示。
- ☐ Total IPv6 addresses: $2^{128} \approx 340$ undecillion. Enough for every atom on Earth to have an IP!
- ☐ Transition challenges: IPv4 and IPv6 coexist. Dual-stack, tunneling, translation.

Activity (10 min): Convert IPv4 address to binary: 172.16.254.1 → ? Calculate: How many /24 subnets in 10.0.0.0/8?

Plenary (5 min): "Why has IPv6 adoption been slow despite IPv4 exhaustion?" Discuss: cost, compatibility, NAT workaround.

Homework: Find your device's IP address (ipconfig / ifconfig). Is it public or private? Explain.

Bilingual Keywords:

- ☐ IP address IP地址 (IP dìzhǐ)
- ☐ Subnet mask 子网掩码 (zǐwǎng yǎnmǎ)
- ☐ NAT 网络地址转换 (wǎngluò dìzhǐ zhuǎnhuàn)
- ☐ IPv6 互联网协议第六版 (hùliánwǎng xiéyì dì liù bǎn)

Lesson 3 (40 min): DNS & Mid-Unit Review 域名系统与单元复习

Learning Objectives:

- ☐ Explain purpose and operation of DNS 解释DNS的目的和运作方式
- ☐ Describe DNS hierarchy 描述DNS层次结构
- ☐ Review Weeks 4–6 content 复习第4–6周内容

Starter (5 min): How does typing "google.com" take you to Google's servers? There are millions of servers!

Main Activity (20 min):

- ☐ DNS (Domain Name System): Translates human-readable domain names to IP addresses. DNS将域名转换为IP地址。
- ☐ Hierarchical structure: Root (.) → TLD (.com, .org, .cn, .edu) → Domain (google) → Subdomain (www).
- ☐ DNS resolution process:
 1. Browser checks cache
 2. OS checks hosts file and cache

3. Query recursive DNS resolver (ISP's DNS server)
4. Resolver queries root server → TLD server → authoritative nameserver
5. IP address returned, cached for TTL (Time To Live)

☐ Record types: A (IPv4), AAAA (IPv6), MX (mail), CNAME (alias), NS (nameserver)

☐ Exam tip: DNS operates at Application Layer of TCP/IP.

Review Activity (10 min): Speed dating — Students rotate, asking each other questions from Weeks 4–6.

Plenary (5 min): "What would happen if all DNS servers stopped working?" Discuss: Internet still works with IP addresses, but usability collapses.

Week 6 Assessment: Mid-Unit Test — Communication (45 minutes)

- ☐ Q1: Network types and models (8 marks)
- ☐ Q2: Network hardware comparison (6 marks)
- ☐ Q3: Packet switching explanation (6 marks)
- ☐ Q4: Protocol identification (5 marks)
- ☐ Q5: TCP/IP stack layers (8 marks)
- ☐ Q6: IP addressing and DNS (7 marks)
- ☐ Total: 40 marks — 10% of semester grade

Week 7: CPU Architecture & the Fetch-Decode-Execute Cycle CPU架构与取指-译码-执行周期

Lesson 1 (40 min): Von Neumann Architecture & CPU Components 冯·诺依曼架构与CPU组件

Learning Objectives:

- ☐ Describe Von Neumann architecture 描述冯·诺依曼架构
- ☐ Identify and explain CPU components 识别并解释CPU组件
- ☐ Explain role of buses 解释总线的作用

Starter (5 min): What does a CPU look like? Show images. What's inside that small chip?

Main Activity (20 min):

☐ Von Neumann architecture (1945): Stored program concept — instructions and data in same memory.
冯·诺依曼架构: 存储程序概念。

☐ Key components:

— ALU (Arithmetic Logic Unit): Performs calculations and logical operations. 算术逻辑单元: 执行计算和逻辑运算。

— CU (Control Unit): Coordinates CPU operations, decodes instructions, sends control signals.
控制单元: 协调CPU操作, 解码指令。

— Registers: Ultra-fast storage locations inside CPU. 寄存器: CPU内部的超高速存储位置。

☐ Buses (data pathways):

— Data bus: Carries data between components. Bidirectional. Width affects performance (32-bit vs 64-bit).
数据总线: 在组件之间传输数据。

— Address bus: Carries memory addresses. Unidirectional (CPU to memory). Width determines max memory (32-bit = 4GB).
地址总线: 传输内存地址。

— Control bus: Carries control signals (read, write, clock, interrupt). 控制总线: 传输控制信号。

☐ Clock: Synchronises operations. Clock speed measured in GHz. Higher = more instructions per second (but also more heat).

Activity (10 min): Label blank CPU diagram. Add function of each component in English + Chinese.

Plenary (5 min): "Why is a 64-bit processor faster than 32-bit for large calculations?" Discuss wider data bus, larger address space.

Homework: Research: What is clock speed of your phone's CPU? Your laptop's? Why do they differ?

Bilingual Keywords:

- ☐ Architecture 架构 (jiàgòu)
- ☐ Bus 总线 (zǒngxiàn)
- ☐ Clock speed 时钟速度 (shízhōng sùdù)
- ☐ Register 寄存器 (jìcúnqì)

Lesson 2 (40 min): The Fetch-Decode-Execute Cycle 取指-译码-执行周期

Learning Objectives:

- ☐ Explain each stage of FDE cycle 解释FDE周期的每个阶段
- ☐ Trace register contents through FDE cycle 追踪FDE周期中的寄存器内容
- ☐ Calculate execution time given clock speed 根据时钟速度计算执行时间

Starter (5 min): CPU executes billions of instructions per second. How does it know WHAT to do and WHERE to find it?

Main Activity (20 min):

- ☐ FDE cycle = how CPU executes one instruction. Repeats continuously while computer is on.

☐ Fetch:

- PC (Program Counter) holds address of next instruction
- Address copied to MAR (Memory Address Register)
- Address placed on address bus
- Control bus sends "read" signal
- Instruction copied from memory to MDR (Memory Data Register)
- Instruction copied to CIR (Current Instruction Register)
- PC incremented by 1 (or instruction length)

☐ Decode:

- CU decodes instruction in CIR
- Determines operation (ADD, LOAD, STORE, JUMP, etc.)
- Determines operands (registers, memory addresses, immediate values)

☐ Execute:

- ALU performs calculation, or data moved between registers/memory
- Result stored in accumulator or designated register
- Status flags updated (zero, negative, overflow, carry)

☐ Worked example: Trace "ADD 5" through FDE cycle. Show register contents at each stage.

☐ Clock cycles: Fetch (1), Decode (1), Execute (1-3+). Total = 3-5 clock cycles per instruction.

☐ At 3 GHz, one cycle = 0.33 nanoseconds. 3 billion+ cycles per second.

Activity (10 min): Students trace "LOAD 10", "ADD 15", "STORE 20" through FDE cycle. Fill trace table.

Plenary (5 min): Exam question: "Describe the role of the Program Counter in the fetch stage." (2 marks)

Homework: Trace table for 3 instructions. Show PC, MAR, MDR, CIR, Acc contents after each stage.

Common Misconceptions:

- ☐ ☐ "PC increments at the end of Execute" — It increments at end of Fetch

- ☐ ☐ "MDR and CIR are the same" — MDR holds data from memory; CIR holds current instruction being decoded
- ☐ ☐ "All instructions take same time" — Complex instructions (division) take more cycles than simple (addition)

Lesson 3 (40 min): Memory Hierarchy & Cache 存储层次结构与缓存

Learning Objectives:

- ☐ Describe memory hierarchy 描述存储层次结构
- ☐ Explain role of cache memory 解释高速缓存的作用
- ☐ Compare RAM, ROM, cache, secondary storage 比较RAM、ROM、缓存和二级存储

Starter (5 min): Why does your computer have RAM, SSD, AND cache? Why not just one big fast memory?

Main Activity (20 min):

- ☐ Memory hierarchy triangle: Fast/small/expensive at top; slow/large/cheap at bottom.
- Level 1 (L1) Cache: On CPU, 32-64 KB, fastest (1-3 ns), stores most frequently used data.
- Level 2 (L2) Cache: On/ near CPU, 256 KB-1 MB, slightly slower.
- Level 3 (L3) Cache: Shared across cores, 4-32 MB.
- RAM (DRAM): 8-64 GB, slower (100 ns), volatile.
- Secondary Storage (SSD/HDD): 256 GB-4 TB+, slowest (microseconds-milliseconds), non-volatile.
- ☐ Principle of locality: Programs tend to access same data/ nearby data repeatedly. Cache exploits this.
- ☐ Cache hit: Data found in cache (fast). Cache miss: Data not in cache (slow, must fetch from RAM).
- ☐ RAM vs ROM:
 - RAM: Volatile, read/write, stores running programs/data. DRAM (cheap, needs refresh), SRAM (fast, expensive, used in cache).
 - ROM: Non-volatile, read-only (mostly), stores BIOS/firmware. PROM, EPROM, EEPROM.
- ☐ Virtual memory: Uses HDD/SSD as extension of RAM when RAM full. Slower but prevents crashes.

Activity (10 min): Design optimal computer for: (a) gaming, (b) server, (c) budget laptop. Justify memory choices.

Plenary (5 min): "Why does adding more RAM often speed up a computer more than upgrading CPU?" Discuss: reduces virtual memory swapping.

Homework: Research: What are L1/L2/L3 cache sizes in Apple M3 chip? Intel i9? Why does Apple use "unified memory"?

Bilingual Keywords:

- ☐ Cache 高速缓存 (gāosù huǎncún)
- ☐ Volatile 易失性 (yìshī xìng)
- ☐ Non-volatile 非易失性 (fēi yìshī xìng)
- ☐ Virtual memory 虚拟内存 (xūnǐ nèicún)

Week 8: Secondary Storage & I/O Devices 二级存储与输入输出设备

Lesson 1 (40 min): Secondary Storage Types 二级存储类型

Learning Objectives:

- ☐ Compare magnetic, optical, and solid-state storage 比较磁存储、光存储和固态存储
- ☐ Explain how each storage technology works 解释每种存储技术的工作原理
- ☐ Select appropriate storage for given scenarios 为特定场景选择合适的存储

Starter (5 min): Show HDD, SSD, USB drive, DVD, SD card. What's inside each? How do they store data?

Main Activity (20 min):

☐ Magnetic storage: HDD. Platters spin, read/write head magnetises sectors. High capacity, low cost, mechanical failure risk. 磁存储: 硬盘驱动器, 盘片旋转, 磁头读写。

☐ Optical storage: CD/DVD/Blu-ray. Pits and lands read by laser. Removable, durable, but lower capacity. Becoming obsolete.

☐ Solid-state storage: SSD, USB flash, SD cards. NAND flash memory (no moving parts). Fast, durable, expensive per GB. 固态存储: 无移动部件, 速度快, 耐用。

☐ Emerging: M.2 NVMe (PCIe interface, very fast), Intel Optane (persistent memory), DNA storage (research).

☐ Comparison table: Speed, capacity, durability, cost, portability, power consumption.

☐ Exam scenario: "A photographer needs to store 10,000 RAW images. Recommend storage." Answer: SSD for active work, HDD for archive, cloud for backup.

Activity (10 min): Group decision task: Choose storage for (a) gaming PC, (b) space probe, (c) hospital records, (d) movie distribution.

Plenary (5 min): "Why are SSDs replacing HDDs in laptops but not in data centres?" Discuss cost/GB, capacity needs, write endurance.

Homework: Calculate cost per GB for HDD vs SSD using current prices. Project 5-year trend.

Common Misconceptions:

☐ ☐ "SSD and USB flash are completely different" — Both use NAND flash; form factor and controller differ

☐ ☐ "Optical discs are dead" — Still used for archival, distribution, regions without broadband

Lesson 2 (40 min): Input & Output Devices 输入与输出设备

Learning Objectives:

☐ Categorise input and output devices 分类输入和输出设备

☐ Explain how sensors and actuators work 解释传感器和执行器的工作原理

☐ Discuss role of I/O in embedded systems 讨论输入输出在嵌入式系统中的作用

Starter (5 min): Your smartphone: List 10 input methods and 5 output methods. Compare with desktop computer.

Main Activity (20 min):

☐ Input devices: Keyboard, mouse, touchscreen, microphone, scanner, barcode reader, NFC, biometric sensors, camera, GPS, accelerometer.

☐ Output devices: Monitor (LCD, OLED), printer (laser, inkjet, 3D), speakers, projector, haptic feedback, LED indicators.

☐ Sensors: Detect physical conditions and convert to electrical signals. Temperature, pressure, light, proximity, gyroscope.

☐ Actuators: Convert electrical signals to physical action. Motors, valves, relays, solenoids.

☐ Embedded systems: Dedicated computer within larger system. Examples: Washing machine, smart thermostat, car ECU, medical monitor.

— Inputs: Sensors detect load, temperature, water level

— Processing: Microcontroller runs control program

— Outputs: Actuators control motor, heater, valves

☐ I/O interfaces: USB, HDMI, Bluetooth, SPI, I2C (embedded).

Activity (10 min): Design embedded system for smart greenhouse. List all sensors, actuators, processing, I/O.

Plenary (5 min): "Why do embedded systems often use microcontrollers instead of full CPUs?" Discuss: cost, power, size, dedicated function.

Homework: Research IoT device (smartwatch, fitness tracker). List all sensors, explain what each measures.

Bilingual Keywords:

- ☐ Sensor 传感器 (chuángǎnqì)
- ☐ Actuator 执行器 (zhíxíngqì)
- ☐ Embedded system 嵌入式系统 (qiànrù shì xìtǒng)
- ☐ Interface 接口 (jiēkǒu)

Lesson 3 (40 min): IGCSE Bridge & Consolidation IGCSE衔接与巩固

Learning Objectives:

- ☐ Review all Weeks 1–8 content 复习第1–8周全部内容
- ☐ Identify links between IGCSE and A-Level depth 识别IGCSE与A-Level深度之间的联系
- ☐ Practise exam-style questions 练习考试风格题目

Starter (5 min): "What did you learn in IGCSE about binary? How is A-Level different?" Brainstorm in groups.

Main Activity (20 min):

- ☐ IGCSE → A-Level progression:
 - IGCSE: Convert binary to denary, basic file sizes
 - A-Level: Two's complement, hex conversions, audio calculations, compression algorithms, detailed FDE cycle
 - IGCSE: Know what CPU, RAM, ROM are
 - A-Level: FDE cycle with registers, cache hierarchy, bus widths, memory management
 - IGCSE: Basic network types
 - A-Level: TCP/IP stack, packet switching, DNS resolution, IPv4/IPv6
- ☐ Exam question practice: 6 questions from past papers (2 marks each). Peer marking with mark scheme.
- ☐ Error analysis: Common mistakes from diagnostic test reviewed.

Plenary (10 min): Student-led: Each student shares one thing they understand better now and one thing still confusing.

Homework: Create personal revision notes for Weeks 1–8. Include bilingual glossary, key formulas, common misconceptions.

EAL Support:

- ☐ Provide sentence frames for explaining concepts: "____ is important because ____"
- ☐ Word bank for exam answers: "converts, represents, stores, calculates, compares, explains"
- ☐ Allow Chinese annotations in revision notes (not in exam, but aids understanding)

Week 9: Half-Term Assessment & Review 期中评估与复习

Half-Term Examination (Week 9)

Format: Paper 1 Section A style — 1 hour, 35 marks

Section A: Short Answer (20 marks)

- ☐ Q1: Number conversions (binary ☐ denary ☐ hex) (4 marks)
- ☐ Q2: Two's complement representation (3 marks)
- ☐ Q3: File size calculations (image + audio) (5 marks)
- ☐ Q4: Compression method selection and justification (4 marks)
- ☐ Q5: Network hardware identification (4 marks)

Section B: Structured Questions (15 marks)

- ☐ Q6: Packet switching explanation with diagram (5 marks)
- ☐ Q7: TCP/IP layers — match protocols and explain functions (5 marks)

☐ Q8: FDE cycle trace table (5 marks)

Marking: Teacher-marked with detailed feedback. Students complete reflection sheet identifying 3 strengths and 3 areas for improvement.

Post-Assessment:

☐ W9 L2: Review lesson — Go through each question, address common errors

☐ W9 L3: Targeted intervention — Students grouped by weakness area for focused support

☐ 考后反思: 学生完成反思表, 识别优势和改进领域。

Week 9 Lesson Plan: Assessment Review & Intervention

Lesson 1 (40 min): Examination

☐ Formal exam conditions

☐ Seating plan by register

☐ Equipment: Pen, pencil, ruler, calculator (non-programmable)

Lesson 2 (40 min): Whole-Class Review

☐ Display mark scheme

☐ Students self-assess using green pen

☐ Teacher highlights common errors:

— Forgetting to convert bits to bytes

— Confusing MAR and MDR

— Missing "packets can take different routes" in packet switching

— Not justifying compression choice with context

☐ Model answers for 5-mark questions

Lesson 3 (40 min): Differentiated Intervention

☐ Group A (below 50%): Reteach core concepts with visual aids, simplified language, extra practice

☐ Group B (50 - 75%): Targeted practice on weak areas, exam technique focus

☐ Group C (above 75%): Extension work — A2 preview topics, complex problem solving

☐ 分层干预: 根据成绩分组进行针对性辅导

Weeks 1-9 Common Misconceptions Summary

1. Binary conversions: Students forget place values or confuse 2^7 with 2^8 . Use place-value grid.

2. Two's complement: "Leftmost bit is just sign" — No, it has negative weight. Use number line visual.

3. File size calculations: Mixing bits and bytes. Emphasise: 8 bits = 1 byte. Always check units.

4. Compression: "Lossy damages the original" — Compressed copy is separate. Original untouched.

5. TCP/IP: "TCP and IP are the same" — TCP = transport (reliability); IP = network (addressing).

6. FDE cycle: PC increments after Fetch, not after Execute. Use timeline diagram.

7. Cache: "More cache always better" — Diminishing returns, cost, power consumption matter.

8. DNS: Confusing domain names with IP addresses. DNS translates between them.

Address each misconception explicitly in lessons with counter-examples and visual models.

Victoria Park Academy Shenzhen — A-Level Computer Science Curriculum 2024 - 2026